

BACHELOR THESIS

Olivier Amacker

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Numerical Stability of Functional MRI Connectivity Biomarkers



Informatique et Systèmes de communication (ISC)

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University of Applied Sciences and Arts Western Switzerland

Faculty of Engineering and Architecture

Haute École d'Ingénierie de Sion

Olivier Amacker

Numerical Stability of Functional MRI Connectivity Biomarkers

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
Haute École d'Ingénierie de Sion

Supervisor: **PROF. DR OSCAR ESTEBAN**

Co-supervisor: **DR OKITO YAMASHITA**

Expert: **DR AYUMU YAMASHITA**

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PROJECT REPOSITORY

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Abstract

- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear, and usually does not exceed 250–300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

TODO Double check

As [functional Magnetic Resonance Imaging](#) (fMRI) pipelines become more computationally heavy, requiring high-dimensional computations, it is crucial to assess how small numerical variability introduced by several different factors like the running [Operating System](#) (OS), parallelization strategies, and hardware architecture affects the results of these pipelines.

Building upon the paper “Numerical Variability of functional [Magnetic Resonance Imaging](#) (MRI) Graph Measures”^[1], this work goes a step further than looking at different MRI graph measures by also looking at [Major Depressive Disorder](#) (MDD) biomarkers, specifically the [Principal Component analysis](#) (PCA) based biomarkers proposed by the paper “Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection”^[2] by perturbing several steps of the pipeline. It also assesses the impact of numerically perturbed fMRIPrep preprocessing on FC Functional Connectivity (FC) matrices using the Numerical Population Variability Ratio (NPVR) metric.

To reproduce the results of the original study^[1], the results were obtained by running the fMRIPrep^[3] (25.2.5) preprocessing pipeline through a [custom Docker container](#) with floating-point arithmetic perturbations introduced by the Verificarlo^[4] and Fuzzy^[5] libraries. After preprocessing, FC matrices were obtained using Nilearn^[6] (0.13.1). Different absolute thresholds were applied to the FC matrices, and graph metrics were computed using NetworkX^[7] (3.6.1).

For the assessment of the [PCA](#) feature selection, two steps of the pipeline were perturbed using a [specific Fuzzy Docker image](#), the correlation coefficient computation and principal component analysis.

The results given by the original work reproduction closely matched the original ones, except for a specific graph where the Y-axis values were hardcoded inside the original code. The [PCA](#) feature extraction appeared to be very stable under the introduced numerical perturbations, simulating [OS](#) level numerical variability. Giving the exact same results.

These findings suggest that the PCA features extraction is not only stable across different imaging sites and datasets, but also robust to numerical variability.

Keywords

medical data science, fMRI, fMRIPrep, numerical stability, benchmark, PCA, neuro imaging



Résumé

TODO Double check

Alors que les pipelines d'[imagerie par résonance magnétique fonctionnelle](#) (IRMf) deviennent de plus en plus lourdes computationnellement, nécessitant des calculs en haute dimension, il est crucial d'évaluer comment de petites variations numériques introduites par plusieurs facteurs tels que le [Système d'Exploitation](#) (SE) utilisé, les stratégies de parallélisation et l'architecture matérielle affectent les résultats de ces dernières.

S'appuyant sur le papier « Numerical Variability of functional [Imagerie par Résonance Magnétique](#) (IRM) Graph Measures » ^[1], ce travail va plus loin que l'examen de différentes mesures de graphes d'IRM en examinant également les biomarqueurs du Trouble Dépressif Majeur (TDM), notamment les biomarqueurs basés sur l'[Analyse en Composantes Principales](#) (ACP) proposés par le papier "Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection" ^[2], en perturbant plusieurs étapes du pipeline. Il évalue également l'impact du prétraitement fMRIPrep numériquement perturbé sur les matrices de Connectivité Fonctionnelle (CF) en utilisant la métrique Numerical Population Variability Ratio (NPVR).

Pour reproduire les résultats de l'étude originale ^[1], les résultats ont été obtenus en exécutant le pipeline de prétraitement fMRIPrep ^[3] (25.2.5) dans un [conteneur Docker personnalisé](#) avec des perturbations arithmétiques en virgule flottante introduites par les bibliothèques Verificarlo ^[4] et Fuzzy ^[5]. Après prétraitement, les matrices de CF ont été obtenues à l'aide de Nilearn ^[6] (0.13.1). Différents seuils absolus ont été appliqués aux matrices de CF, et les métriques de graphes ont été calculées à l'aide de NetworkX ^[7] (3.6.1).

Pour l'évaluation de la sélection de caractéristiques par ACP, deux étapes du pipeline ont été perturbées à l'aide d'une [image Docker Fuzzy spécifique](#) le calcul des coefficients de corrélation et l'analyse en composantes principales.

Les résultats de la reproduction de l'étude originale correspondent étroitement aux résultats originaux, à l'exception d'un graphe spécifique dont les valeurs de l'axe Y étaient codées en dur dans le code original. L'extraction de caractéristiques par ACP s'est avérée très stable sous les perturbations numériques introduites, simulant la variabilité numérique au niveau du SE, tout en produisant des résultats strictement identiques.

Ces résultats suggèrent que l'extraction de caractéristiques par ACP est non seulement stable entre différents sites d'imagerie et jeux de données, mais également robuste face à la variabilité numérique.

Mots-clés

medical data science, fMRI, fMRIPrep, numerical stability, benchmark, PCA, neuro imaging

Acknowledgements

The **Acknowledgements** section of a bachelor thesis is where you express gratitude to those who supported you during your research and writing process. It is an **OPTIONAL** section. It may include:

- Academic supervisors or advisors who provided guidance.
- Professors or instructors who offered feedback or resources.
- Family and friends for emotional or practical support.
- Institutions or organizations that provided funding, facilities, or data.
- Anyone else who contributed significantly to your work.

Keep this section concise and sincere. It is typically placed after the abstract and before the main content of your thesis.

I want to thank the following people and institution without which played a huge role in making this Bachelor Thesis possible:

- Prof. Dr. Oscar Esteban, for giving me the opportunity to conduct this Bachelor Thesis abroad, as well as for his guidance and supervision.
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- My family and friends back home, for their emotional support and encouragement during this journey.

Chapter 1

Writing a thesis

Writing a report is an exercise that involves both **content** and **form**. In this document, we aim to simplify the formatting aspect without making any assumptions about the content, specifically in the context of the ISC degree programme¹.

1.1 – The content of a thesis

The general structure of a bachelor thesis typically includes the following sections:

1. **Abstract:** A concise summary of the thesis, including the research question, methodology, results, and conclusions.
2. **Résumé:** A summary of the thesis in French.
3. **Acknowledgements:** [Optional] A section to thank those who supported your work.
4. **Table of Contents:** An organized list of chapters and sections.
5. **Introduction:** Presents the background/context, motivation, objectives, and scope and plan of the thesis.
6. **State of the Art / Literature Review:** Reviews existing research and situates the thesis within the academic context, if relevant to your work.
7. **Development and Methodology:** Describes the methods, materials, and procedures used in the research/thesis.
8. **Results:** Presents the findings of the research, often with tables, figures, and analysis.
9. **Discussion:** Interprets the results, discusses implications, and relates findings to the research question.
10. **Conclusion:** Summarizes the main findings and contributions, and suggests future work.
11. **References / Bibliography:** Lists all sources cited in the thesis.
12. **Appendices:** (Optional) Contains supplementary material such as raw data, code, or additional explanations.

This structure may vary depending on the field of study, but these elements are commonly found in most bachelor theses. They are recommended for the *ISC Bachelor thesis* and should be adapted to the specific requirements of your thesis (e.g., if you have a state of the art section or not).

You can also change the order or the names of the sections, for instance, if you want to put the state of the art before the introduction, or if you want to add a section on methodology before the results.

¹ Here is how to add a footnote <https://isc.hevs.ch>

1.2 – Academic titles

Please note that the academic titles of your supervisors and experts are important.

They should be included on the cover page, and you should use the correct title when addressing them in the acknowledgements section. For instance, a professor should be addressed as “Prof. [Name]”, while a doctor should be addressed as “Dr [Name]” (**without a colon!**). A professor who is also a doctor should be addressed as “Prof. Dr [Name]”.

If you are unsure about the title of your supervisor, co-supervisor, or expert, you can ask them directly or check their profile on the university website.

1.3 – Compiling the thesis

If you compile your thesis using the `typst` command line tool, or by using the `typst` extension in Visual Studio Code, please note that you must install the fonts used in this template. You can do so by running the following command in your terminal:

```
./fonts/install_fonts.sh
```

Chapter 2

Introduction

Have fun **TODO** writing your thesis! and good luck with it !



Figure 1 - Grace Hopper

Chapter 3

Development and Methodology 1

Chapter 4

Development and Methodology 2

Chapter 5

Development and Methodology 3

Chapter 6

Development and Methodology 4

Chapter 7

Development and Methodology 5

Chapter 8

Development and Methodology 6

Chapter 9

Development and Methodology 7

Chapter 10

Results and Discussion

Chapter 11

Conclusion

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Appendix

Table of Acronyms

ACP :	Analyse en Composantes Principales
ATR :	Advanced Telecommunications Research Institute International
CF :	Connectivité Fonctionnelle
FC :	Functional Connectivity
IRM :	Imagerie par Résonance Magnétique
IRMf :	imagerie par résonance magnétique fonctionnelle
MDD :	Major Depressive Disorder
MRI :	Magnetic Resonance Imaging
NPVR :	Numerical Population Variability Ratio
OS :	Operating System
PCA :	Principal Component analysis
PR :	Pull Request
SE :	Système d'Exploitation
TDM :	Trouble Dépressif Majeur
fMRI :	functional Magnetic Resonance Imaging

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Code Appendix

```

1  class Cons[+A](hd: A, tl: ⇒ MyStream[A]) extends MyStream[A] {
2      override def isEmpty: Boolean = false
3      override val head: A = hd
4      override lazy val tail: MyStream[A] = tl
5      override def #::[B >: A](elem: B): MyStream[B] =
6          new Cons[B](elem, this)
7
8      override def ++[B >: A](anotherStream: ⇒ MyStream[B]): MyStream[B] =
9          new Cons[B](head, tail ++ anotherStream)
10
11     override def foreach(f: A ⇒ Unit): Unit = {
12         f(head)
13         tail.foreach(f)
14     }
15 }

```

Listing 1 - Code included from the file example.scala

```

1  def mergeSort(arr):
2      if len(arr) ≤ 1:
3          return arr
4
5      mid = len(arr) // 2
6      leftHalf = arr[:mid]
7      rightHalf = arr[mid:]
8
9      sortedLeft = mergeSort(leftHalf)
10     sortedRight = mergeSort(rightHalf)
11
12     return merge(sortedLeft, sortedRight)
13
14 def merge(left, right):
15     result = []
16     i = j = 0
17
18     while i < len(left) and j < len(right):
19         if left[i] < right[j]:
20             result.append(left[i])
21             i += 1
22         else:
23             result.append(right[j])
24             j += 1
25
26     result.extend(left[i:])
27     result.extend(right[j:])
28
29     return result
30
31 mylist = [3, 7, 6, -10, 15, 23.5, 55, -13]
32 mysortedlist = mergeSort(mylist)
33 print("Sorted array:", mysortedlist)

```

Listing 2 - Second code included from the file example.scala

```

1  class Cons[+A](hd: A, tl: ⇒ MyStream[A]) extends MyStream[A] {
2      override def isEmpty: Boolean = false
3      override val head: A = hd
4      override lazy val tail: MyStream[A] = tl

```

```
5  override def #::[B >: A](elem: B): MyStream[B] =  
6      new Cons[B](elem, this)  
7  
8  override def ++[B >: A](anotherStream: ⇒ MyStream[B]): MyStream[B] =  
9      new Cons[B](head, tail ++ anotherStream)  
10  
11  override def foreach(f: A ⇒ Unit): Unit = {  
12      f(head)  
13      tail.foreach(f)  
14  }  
15 }
```

Listing 3 - Second code included from the file example.scala

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